Unsupervised Hyperspectral and Multispectral Image Fusion With Deep Spectral–Spatial Collaborative Constraint

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***Abstract*— The most cost-effective way to obtain a high spatial resolution hyperspectral image (HrHSI) is to fuse a low spatial resolution hyperspectral image (LrHSI) and corresponding high spatial resolution multispectral image (HrMSI). This article proposes a generalizable unsupervised deep fusion method based on spectral–spatial collaborative constraint to address LrHSI and HrMSI fusion task. First, in view of the limitations of the current spectral–spatial downsampled model, the group convolution enhancement (GCE) module is designed to eliminate the radiometric difference between the images to be fused. Second, to enhance the model's feature extraction ability, this article introduces the design of the spatial, channel, and filter 3-D attention factor dynamic convolutional kernel (SCFConv). In order to verify the proposed method, we compared and evaluated our method with traditional methods and unsupervised deep learning methods using both simulated and real onboard data, respectively. In the absence of HrHSI validation images in real scenarios, we evaluate the performance of different fusion models through classification results. The experimental results demonstrate the effectiveness of the proposed model and the practical value of the fusion results (the onboard data produced by ours are available at https://drive.google.com/drive/folders/1JLCCB6ld5R49HDLN5SsMISx1d0fuqRjO).**

***Index Terms*— Dynamic convolution, hyperspectral and multispectral image fusion, hyperspectral image classification, unsupervised deep learning, ZY-1(02D).**

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I. Introduction

HYPERSPECTRAL images with low spatial resolution hyperspectral image (LrHSI) typically consist of numerous spectral bands. As a result of constraints inherent to the imaging sensor, LrHSI is often due to the limited spatial resolution, which can lead to challenges such as mixing of image elements and blurred boundaries. These limitations significantly restrict the potential applications and spectral advantages of LrHSI. In contrast, high spatial resolution multispectral images (HrMSIs) are characterized by distinct texture features but may lack distinctive spectral features. Distinguishing complex features solely based on limited channel information proves challenging [1]. Therefore, obtaining high spatial resolution hyperspectral images (HrHSIs) is of great research significance for various applications in remote sensing, such as classification [2], [3], [4], [5], [6], [7] and target detection [8], [9], [10], [11], [12], [13]. Currently, the more prevalent research approach involves obtaining HrHSI through the fusion of HrMSI and LrHSI with the same coverage. The related methods can be divided into traditional methods and deep learning-based fusion methods.

Traditional methods can be classified into the following categories, including methods based on pan-sharpening extension [14], [15], [16], Bayesian-based methods [17], [18], [19], matrix factorization-based methods [20], [21], [22], and tensor representation-based methods [23], [24], [25].

The HrMSI and LrHSI fusion methods based on pan-sharpening extension are developed on the principle of HrMSI and panchromatic (PAN) fusion. Aiazzi et al. [26] split the problem of LrHSI and HrMSI fusion into several subproblems similar to the PAN and HrMSI fusion. Each subproblem is the fusion of a single band of the HrMSI with multiple bands of the LrHSI. Vivone et al. [27] provided a detailed introduction and comparison of methods for pan-sharpening extension. This type of method is computationally efficient, but it is difficult to ensure that the spatial texture information and spectral reconstruction achieve better results simultaneously. Especially when the gap between the spatial resolutions of the images to be fused is large, in order to recover the spatial information, it will lead to spectral distortions.

Regarding Bayesian-based methods, they are utilized for estimating the target image by typically incorporating prior regularity into the observation model and leveraging the

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posterior probability. This approach improves the accuracy and robustness of the estimate. Obviously, this type of method overly relies on the accuracy of priori information, which limits the development of the model.

The matrix factorization-based methods primarily involve principal component analysis (PCA) and nonnegative matrix factorization (NMF). These methods achieve fusion by decomposing LrHSI and HrMSI into low-rank and sparse matrices. One advantage of these methods is that they have a better performance. However, their performance depends on the parameter selection and has a high computation cost.

The tensor representation-based methods primarily consist of two types, namely, Tucker decomposition and canonical polyadic (CP) decomposition. These methods represent LrHSI and HrMSI as the product of the kernel tensor and the factor matrix. The fusion results are obtained by combining these decomposition results. One advantage of these methods is that they can capture the nonlinear relationship between LrHSI and HrMSI. Therefore, they are suitable for complex data. However, the disadvantage is that the computational complexity is high, and it requires many computational resources.

In recent years, deep learning has been widely used in the field of LrHSI and HrMSI fusion due to its powerful feature extraction capabilities [28]. Such methods can be specifically categorized into supervised and unsupervised. Zhao et al. [29] introduce an innovative GAN framework with a two-stage adversarial training strategy and a spatial–spectral attention block to solve the fusion task. To tackle the challenges in multisource image fusion, Cao et al. [30] introduce an innovative supervised diffusion model with coarse-grained style modulation and fine-grained wavelet modulation, demonstrating superior performance on fusion benchmarks. The supervised approach requires HrHSI as training samples, which do not exist in real situations. Therefore, it is difficult to apply in practice [31], [32], [33]. Therefore, unsupervised approaches have become a hot spot in current research. Zheng et al. [34] proposed HyCoNet, an unsupervised deep learning-based fusion method, which can solve the problems in LrHSI and HrMSI fusion without a priori information such as point spread function (PSF) and spectral response function (SRF). Zhan et al. [35] proposed an LrHSI and HrMSI fusion method with a new cross-scale octave network, which can extract multiscale spatial information with a progressive image fusion structure.

In recent years, fusion methods based on the spectral–spatial downsampled concept have shown promising performance when transformed into a spectral mapping problem. Han et al. [36] first proposed the spectral–spatial downsampled concept, which converts fusion tasks into spectral mapping problems under the assumption that the PSF and SRF are known. Chen et al. [37] further designed unsupervised fusion models to learn the spectral mapping relationship between LrMSI and LrHSI at the pixel level. Li et al. [38] designed multiple deep learning modules to adaptively learn the PSF and SRF during the degradation process.

However, on the one hand, current methods based on the spectral–spatial degradation approach tend to have overly idealized model designs during the degradation process, overlooking the radiometric differences between the data to be fused. Constrained by imaging time, atmospheric conditions, imaging equipment, and image preprocessing, real LrHSI and HrMSI inputted into the fusion process often exhibit differences in radiometric.

On the other hand, most current unsupervised deep fusion models are performed only on simulation datasets and lack validation on real datasets. In addition, validation images for fusion results are not easily available in practical situations, so there is an urgent need to design evaluation metrics that do not require validation images. At the same time, the purpose of fusion is to combine advantageous information and make a better decision in a specific application. The application-related metrics can be used to evaluate the performance of the fusion results, such as classification and target detection. Taking classification as an example, if the fusion result improves the classification result compared with the input HrMSI, it indirectly proves the significance of the fusion. If the classification accuracy of the HrHSI obtained by the proposed method is superior to other fusion methods, it demonstrates the effectiveness of the proposed method.

Based on the above problems, this article proposes a fusion model of unsupervised deep spectral–spatial collaborative constraint (UDS^{2}C^2). The main contributions of this article are as follows.

1) A 3-D dynamic convolution, SCFConv, is designed to enhance the feature extraction capability of the convolutional neural networks (CNNs) by calculating the attention factor.
2) Considering the radiometric difference between LrHSI and HrMSI, the group convolution enhancement (GCE) module is designed to improve degradation performance.
3) Due to the lack of validation images for real data, we introduce a user-oriented quality measure by evaluating the fusion performance through classification accuracy.

The remainder of this article is organized as follows. Section II introduces the related work on dynamic convolution. Section III describes the design ideas and limitations of the current spectral–spatial downsampled model. Section IV describes the proposed method in detail. In view of the shortcomings of current research, a UDS^{2}C^2 fusion model has been developed. In Section V, the performance of the proposed model is demonstrated through a detailed evaluation. The experiment has both simulation data and ZY-1(02D) real onboard data. Section VI summarizes the overall contribution of this article.

II. Related Work

A. Dynamic Convolution

Single static convolution is a commonly used convolution operation in traditional CNN, which has a fixed convolution kernel, and its main advantages are simple and efficient. However, the design of fixed convolution limits the expressive ability of the model.

Different from traditional single static convolution, dynamic convolution utilizes a set of convolution kernels and

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dynamically adjusts the weight of each kernel based on the input data's features. This is achieved by introducing an attention mechanism, which enhances the model's ability to extract and express features. When processing different types of data, adaptive deallocation of weights improves the accuracy of the model. Yang et al. [39] and Chen et al. [40] calculate the attention weight of the convolution kernel based on the modified SE structure proposed in CondConv and DyConv. For example, CondConv uses the sigmoid function as the activation function to calculate the attention weight, while DyConv uses the softmax function.

B. ODConv

However, the above dynamic convolution method has some limitations. First, the attention mechanism is designed to consider only the dimension of the number of convolution kernels while ignoring the other 3-D of the convolution kernel (space size, number of input channels, and number of output channels). This limits the ability of dynamic convolution to adequately capture rich contextual information. In addition, applying dynamic convolution to multiple convolutional layers significantly increases the number of parameters of the model.

To solve these problems, Li et al. [41] propose a new dynamic convolution design method called omnidimensional dynamic convolution (ODConv). ODConv utilizes a novel multidimensional attention mechanism to learn four types of attention weights in 4-D of the convolutional kernel space. Experiments show that these four types of attention weights complement each other. The gradual application of these four types of attention weights to the corresponding convolution kernels can significantly enhance the feature extraction ability of convolution operations and the model.

ODConv can be used as a plug-in design to replace the regular convolutional layers found in many CNN architectures. Compared with existing dynamic convolutional designs, ODConv achieves a better performance. In addition, the performance improvement of ODConv on the pretrained classification model can also be well transferred to downstream tasks such as object detection, which verifies its good generalization ability.

III. Problem Formulation

The LrHSI $\mathbf{X} \in \mathbb{R}^{wh\times C}$ and the HrMSI $\mathbf{Y} \in \mathbb{R}^{WH\times C_m}$ as input of this model, where w and h represent the width and height of LrHSI, respectively, and C is the number of spectra of LrHSI. W and H represent the width and height of HrMSI, respectively, and C_m is the number of spectra of HrMSI ($w \ll W$, $h \ll H$, and $C_m \ll C$). Then, HrHSI can be represented as $\mathbf{Z} \in \mathbb{R}^{WH\times C}$.

The main principle of learning the spectral upsampling module (as shown in Fig. 1) based on the spectral–spatial downsampling model to generate LrMSI, and then leveraging the LrMSI along with the target LrHSI to learn the spectral upsampling module, is as follows.

Suppose that LrHSI $\mathbf{X}$ and the HrMSI $\mathbf{Y}$ are obtained from the HrHSI $\mathbf{Z}$ through spatial downsampling and spectral downsampling processes

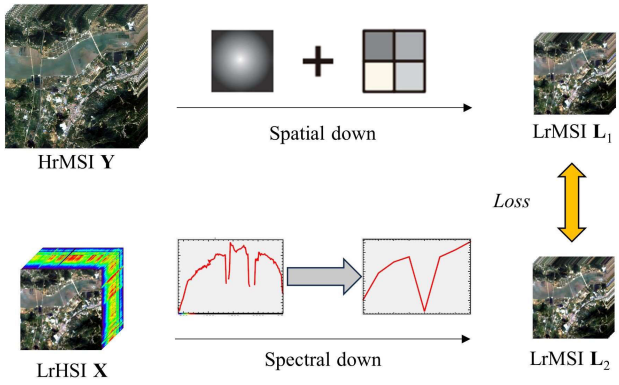


Fig. 1. Illustration of spectral–spatial downsampled model.

$$\mathbf{X} = \mathbf{PZ} \tag{1}$$

$$\mathbf{Y} = \mathbf{ZS}. \tag{2}$$

Among them, $\mathbf{P} \in \mathbb{R}^{wh\times WH}$ and $\mathbf{S} \in \mathbb{R}^{C\times C_m}$ represent the spatial downsampling module and spectral downsampling module, respectively.

By applying the same spectral downsampling operation $\mathbf{S}$ to the LrHSI $\mathbf{X}$ to obtain the LrMSI $\mathbf{L}_1 \in \mathbb{R}^{wh\times Cm}$

$$\mathbf{L}_1 = \mathbf{XS}. \tag{3}$$

Similarly, by applying the same spatial downsampling operation $\mathbf{P}$ to the HrMSI $\mathbf{Y}$ to obtain the LrMSI $\mathbf{L}_2 \in \mathbb{R}^{wh\times Cm}$

$$\mathbf{L}_2 = \mathbf{PY}. \tag{4}$$

Since both LrMSIs are obtained from the same HrHSI $\mathbf{Z}$ through the same spatial and spectral downsampling processes, they are identical. The specific derivation is shown as follows:

$$\mathbf{L}_1 = \mathbf{XS} = \mathbf{PZS} = \mathbf{PY} = \mathbf{L}_2. \tag{5}$$

From (2) and (5), it can be observed that LrMSI $\mathbf{L}_1$ and HrMSI $\mathbf{Y}$ are obtained from LrHSI $\mathbf{X}$ and LrMSI $\mathbf{Z}$, respectively, through the same spectral downsampling operation $\mathbf{S}$. Therefore, the spectral upsampling model learned from LrMSI $\mathbf{L}_1$ and LrHSI $\mathbf{X}$ and the upsampling of HrMSI $\mathbf{Y}$ to HrHSI $\mathbf{Z}$ are identical.

However, the existing spectral–spatial downsampled model has limited effectiveness in the application process. Because they overlook the diversity and complexity of real data. To this end, we enhanced this model and the proposed model structure is as follows.

IV. Proposed Approach

The model proposed in this article, as shown in Fig. 2, is a UDS^{2}C^2 fusion model. It combines the information from a set of LrHSI and HrMSI to obtain HrHSI. Among them, the input images have the same coverage area. An enhanced spectral–spatial downsampled model was used to degrade this set of images to LrMSI by collaborative constraints in both spectral and spatial aspects. Then, during the training stage, the spectral upsampling module is adaptively learned using the generated LrMSI and the original input LrHSI.

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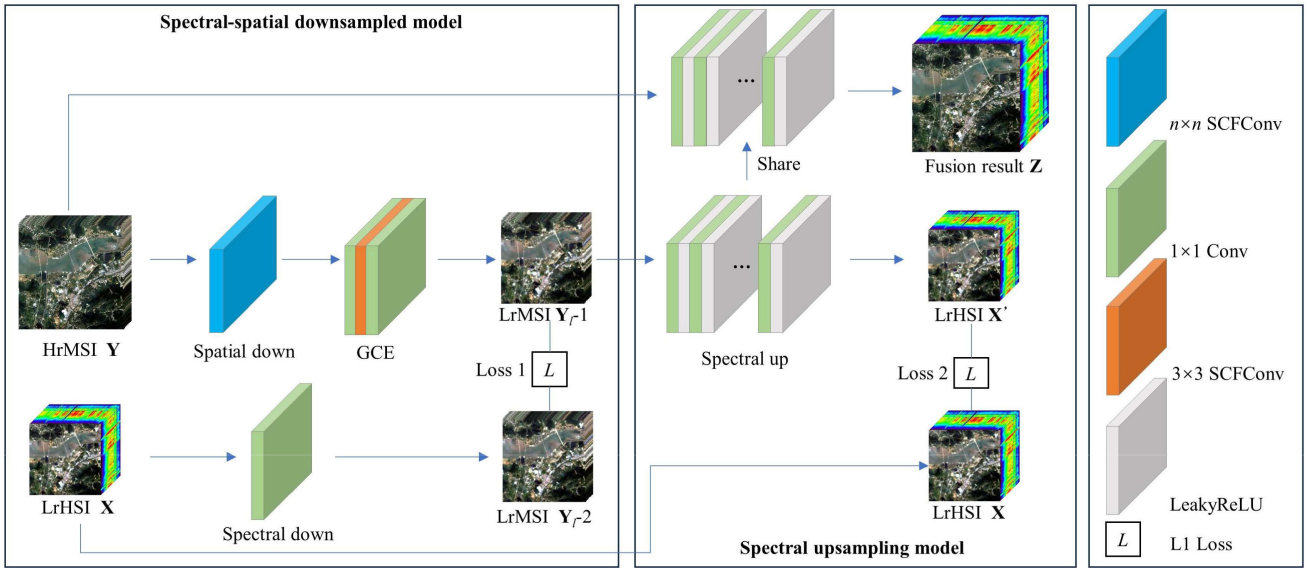


Fig. 2. Illustration of UDS^{2}C^2 fusion model.

This upsampling module is applied to the original input HrMSI in order to obtain the fusion result.

In order to further improve the accuracy of the fusion results, this article uses two stages to train the model. The first stage uses a larger learning rate and a larger number of training sessions to obtain a preliminary fusion result. The second stage uses a smaller learning rate and fewer training cycles to get a more accurate fusion result.

The model generally has high accuracy and stability in LrHSI and HrMSI fusion and can be widely used in practical applications.

A. Enhanced Spectral–Spatial Downsampled Model

The LrHSI $\mathbf{X} \in \mathbb{R}^{wh \times C}$ and the HrMSI $\mathbf{Y} \in \mathbb{R}^{WH \times C_m}$ as input of this model. At this time, the number of spectra and the spatial resolution of $\mathbf{X}$ and $\mathbf{Y}$ are different, so we design the enhanced spectral–spatial downsampled model to turn both $\mathbf{X}$ and $\mathbf{Y}$ into LrMSI.

First, the HrMSI $\mathbf{Y}$ goes through the spatial downsampling module, and the physical meaning of this module is to learn PSF. This module is mainly composed of $n \times n$ SCFConv (for a detailed introduction to SCFConv, see Section IV-C). The number of input and output channels is C_m, and it keeps the number of channels in the image unchanged. The stride parameter is also set to n, where $n = W/w = H/h$ represents the multiple of the difference in spatial dimensions between the LrHSI and the HrMSI.

Then, in this article, in order to reduce the radiometric difference between the images to be fused, the GCE module is added in the spatial downsampling module. Some convolution kernels of the GCE module are SCFConv, and grouped convolution is taken, with the grouping parameter group set to C_m. Grouped convolution divides features into several groups, performs independent convolution within each group, and then stitches all the results together. Group convolution can effectively reduce the amount of calculation and parameters of the model. It can improve the performance of the model.

The GCE module mainly consists of three layers, with the input channel, output channel, and group parameters of each layer denoted as C_m. The first and third layers have the same structure, with a convolution kernel size of 1×1 and no padding. The second layer has a convolution kernel size of 3×3 SCFConv and a padding of 1. The GCE module further optimizes the downsampling process through mapping transformation to reduce radiometric differences and obtain more accurate LrMSI.

Using the powerful ability of dynamic CNN to extract features, we can get the LrMSI $\mathbf{Y}_l - 1 \in \mathbb{R}^{wh \times C_m}$ with spatial dimension (w, h) as follows:

$$\mathbf{Y}_l - 1 = \mathbf{P}'\mathbf{Y}. \tag{6}$$

Among them, $\mathbf{P}' \in \mathbb{R}^{wh \times WH}$ is the spatial downsampling module and GCE.

On the other side, the LrHSI $\mathbf{X}$ goes through the spectral downsampling module. The physical meaning of this module is to learn SRF. This module is mainly composed of many 1×1 Conv. The main effect of 1×1 Conv is to reduce the number of channels of the input image. Progressively, certain bands in the LrHSI are downsampled to one band. The design idea of this module is mainly derived from SRF, which is used to illustrate the optical properties of the satellite sensor, related to the structure and materials of the satellite sensor, and so on. It is generally obtained by laboratory spectral calibration of the remote sensor before launch.

The LrMSI $\mathbf{Y}_l - 2 \in \mathbb{R}^{wh \times C_m}$ obtained after going through the spectral downsampling module. The process is as follows:

$$\mathbf{Y}_l - 2 = \mathbf{X}\mathbf{S}'. \tag{7}$$

Among them, $\mathbf{S}' \in \mathbb{R}^{C \times C_m}$ is the spectral downsampling module.

The above is the structure of the enhanced spectral–spatial downsampled model proposed in this article, using $L1$-norm

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loss as the LrMSI $\mathbf{Y}_l - 1$ and $\mathbf{Y}_l - 2$ loss functions to complete the training of the model as follows:

$$\text{Loss1}(\mathbf{Y}_l - 1, \mathbf{Y}_l - 2) = \frac{1}{whsC_m}\|\mathbf{Y}_l - 1 - \mathbf{Y}_l - 2\|_1. \quad (8)$$

B. Spectral Upsampling

Based on the enhanced spectral–spatial downsampled model, a more accurate set of LrMSI $\mathbf{Y}_l - 1$ and $\mathbf{Y}_l - 2$ is obtained with the same LrHSI $\mathbf{X}$ and HrMSI $\mathbf{Y}$ covering the same area as the input, and parameters of the model are also learned.

At this time, the spatial dimension and resolution of LrMSI $\mathbf{Y}_l - 1 \in \mathbb{R}^{wh \times C_m}$ and LrHSI $\mathbf{X} \in \mathbb{R}^{wh \times C}$ are the same, only the number of channels is different. Thus, the spectral upsampling module is designed in this article. This module consists of several 1×1 convolution kernels, and the number of kernels is determined by C_m and C. The first layer of convolution allows C_m to rise to $2C_m$, the second layer of convolution allows $2C_m$ to rise to $4C_m$, and so on until the number of channels of LrMSI $\mathbf{Y}_l - 1$ is close to C. The output channel of the last layer of convolution is set to C. By using this way of gradually increasing the number of channels, it is better to learn the parameters from C_m channels to C channels.

$\mathbf{Y}_l - 1$ goes through the spectral upsampling module to get $\mathbf{X}' \in \mathbb{R}^{wh \times C}$, and the L1-norm loss is used as $\mathbf{X}'$ and $\mathbf{X}$ loss function to complete the training of the model. The specific process is as follows:

$$\mathbf{X}' = \text{SpecUp}(\mathbf{Y}_l - 1) \quad (9)$$

$$\text{Loss2}(\mathbf{X}', \mathbf{X}) = \frac{1}{whC}\left\|\mathbf{X}' - \mathbf{X}\right\|_1 \quad (10)$$

where SpecUp represents the spectral upsampling module.

The proposed method is an unsupervised deep learning fusion model based on a set of input data to be fused. It is trained in two stages with different learning rates and learning epochs. It adaptively learns the enhanced spectral–spatial downsampled model and the spectral upsampling module for this set of input data. Finally, the spectral upsampling module is applied to the HrMSI $\mathbf{Y} \in \mathbb{R}^{WH \times C_m}$ as the input, and boosting the number of channels of $\mathbf{Y}$–C obtains the desired fusion result HrHSI $\mathbf{Z} \in \mathbb{R}^{WH \times C}$. The specific process is as follows:

$$\mathbf{Z} = \text{SpecUp}(\mathbf{Y}). \quad (11)$$

C. SCFConv

In this article, we propose a dynamic convolution kernel called SCFConv, which is based on ODConv. It utilizes a 3-D attention mechanism and a parallel strategy. It can learn the complementary attention of the convolution kernel in the spatial dimension, channel dimension, and filter dimension across the kernel space. In this article, we reorganize the structure of dynamic convolution to enable training on multiple GPUs.

A regular convolutional layer consists of only single convolutional kernel to extract features from the input. For dynamic convolution, it requires the use of a linear combination of many convolutional kernels. These kernels are dynamically weighted by an attention mechanism, which allows the convolutional operation to be dependent on the input. The definition of dynamic convolution can be formulated as follows:

$$\boldsymbol{y} = (\boldsymbol{\alpha}_{w1}\boldsymbol{W}_1 + \cdots + \boldsymbol{\alpha}_{wn}\boldsymbol{W}_n) \cdot \boldsymbol{x} \quad (12)$$

where the input $\boldsymbol{x}$ has dimensions $w \times h \times c_{\text{in}}$ and the output $\boldsymbol{y}$ has dimensions $w \times h \times c_{\text{out}}$. $\boldsymbol{W}$ is the filter weight, a convolutional layer consists of c_{out} filters, $\boldsymbol{W}_i$ is the ith filter weight with dimensions $k \times k \times c_{\text{in}}$, and α is the attention factor.

ODConv utilizes an attention module similar to SE on the attention function $\pi_{wi}(x)$ structure. ODConv uses multicore branching to calculate attention between the spatial dimensions, input channels, output channels, and multiple convolutional kernels. On the contrary, while the number of kernels in the multicore branching improves the final representation of the model, it also greatly increases the number of parameters of the model. When the number of kernels is set to 4, the overall model's number of parameters is close to four times the original number, which reduces the speed of the model.

We improve the ordinary convolution kernel by using a split-head attention structure similar to ODConv. The structure of $\pi_{wi}(x)$ for calculating the 3-D attention factor α_{wi} is shown in Fig. 3.

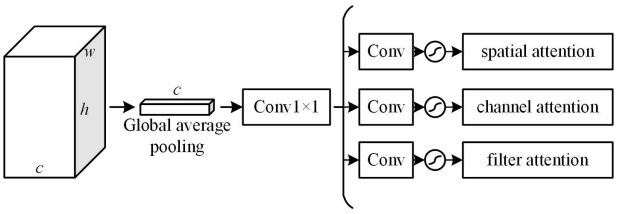


Fig. 3. Structure of $\pi_{wi}(x)$ for computing 3-D attention factor.

Input a feature $\boldsymbol{x} \in \mathbb{R}^{w \times h \times c_{\text{in}}}$, go through a globally averaged pooling layer over the spatial dimensions, and a 1×1 convolutional layer that reduces the number of channels of $\boldsymbol{x}$ to 1/16 of the original one. Then, get the feature $\boldsymbol{f} \in \mathbb{R}^{1 \times 1 \times c_{\text{in}}/16}$ before calculating the SE weights by the ReLU activation function. Finally, the computed SE weights are divided into three branches, after a 1×1 convolutional layer and sigmoid activation function, to compute the spatial attention factor $\boldsymbol{\alpha}_{si} \in \mathbb{R}^{1 \times 1 \times k}$, the channel attention factor $\boldsymbol{\alpha}_{ci} \in \mathbb{R}^{1 \times 1 \times c_{\text{in}}}$, and the filter attention factor $\boldsymbol{\alpha}_{fi} \in \mathbb{R}^{1 \times 1 \times c_{\text{out}}}$, where k denotes the kernel size of an ordinary convolutional kernel, c_{in} denotes the number of input channels, and c_{out} denotes the number of output channels. The resulting attention factor α is finally the Hadamard product with the corresponding dimension of the convolution kernel. SCFConv's structure is shown in Fig. 4.

For lightweight network models, when its depth cannot be expanded, its performance can be improved by replacing the existing convolution with SCFConv.

V. Experimental Results

First, this section describes the configuration of the equipment used in the experiment, including a detailed description

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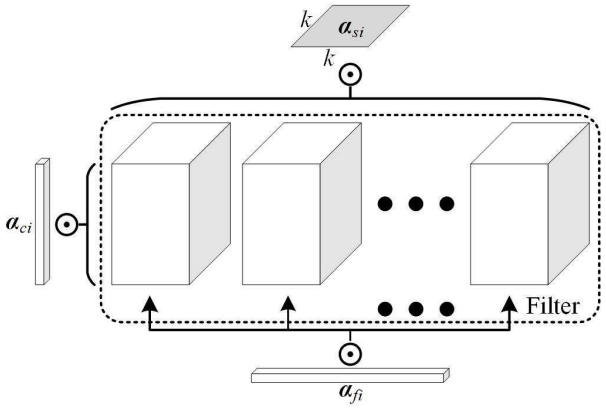


Fig. 4. Three-dimensional dynamic convolution kernel (SCFConv).

of the hardware and software environment. Second, the simulation dataset used in this article and the real onboard dataset processed and produced by this team are introduced. Finally, the proposed method and the current state-of-the-art method are compared under the same evaluation indicators. Through these comparisons, it proves that the fusion model proposed in this article significantly outperforms other methods in LrHSI and HrMSI fusion.

A. Experimental Settings

1) Implementation Details: The proposed fusion method is implemented on PyTorch 1.10.1 and Python 3.7.11. The gradient descent algorithm is trained using Adam. In the first stage, the learning rate and training epoch are set to 0.001 and 30 000, respectively. In the second stage, the learning rate and training epoch are set to 0.00006 and 1000, respectively. The processor in the training environment is an 11th Gen Intel[1] Core[2] i7-11700, and the display adapter is an NVIDIA GeForce RTX 3060.

2) Datasets: In order to validate the generalization of the proposed model, the experiment is not only performed on the simulation dataset Ht18 but also on the actual onboard dataset of two views from ZY-1(02D). This was done to demonstrate the effectiveness and advantages of the proposed method. The corresponding high spatial resolution ground truth reference images are produced by our team for classification-oriented application validation.

a) Ht18: The simulation experiment in this article is based on the Houston2018 dataset, which contains 48 bands covering the wavelength range from 380 to 1050 nm with a high spatial resolution of 1 m. Considering the large spatial extent of the original data, this article uses a data region of size 240 × 240 to facilitate processing. The two bands with noise are removed to simulate the HrHSI, referred to as Ht18. The dimensions of Ht18 are (240, 240, 46), and it is used as the reference image for the fusion results.

To conduct experiments, this article uses the SRF of Worldview 2 to spectrally downsampled Ht18 and obtain an HrMSI with dimensions of (240, 240, 8). At the same time, apply Gaussian filtering and perform spatial downsampling operations. The resulting low spatial resolution LrHSI has dimensions of (48, 48, 46). The dataset is shown in Fig. 5.

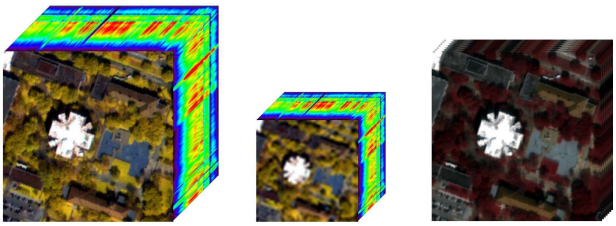


Fig. 5. Assumed HrHSI, LrHSI, and HrMSI after degenerate module processing for Ht18. (a) HrHSI. (b) LrHSI. (c) HrMSI.

b) ZY-1(02D): The ZY-1(02D) satellite is a remote sensing satellite of China, which was successfully launched on December 7, 2019. The satellite carries a hyperspectral payload and a multispectral payload, which can acquire hyperspectral and multispectral images at the same scene. The hyperspectral payload mainly consists of a geostationary hyperspectral imager, which can acquire high-resolution continuous-spectrum remote sensing data. The load can perform LrHSI in the band range from 400 nm to 2500 m, including 76 visible, infrared bands and 90 short-wave infrared bands, totaling 166 bands, with a spatial resolution of 30 m and a width of 60 km. The multispectral load mainly consists of a three-line array multispectral camera, covering eight bands from 452 to 1047 nm, which have a higher spatial resolution (10 m) and 115 km width compared to hyperspectral resolution.

A series of preprocessing on the two sets of raw data from ZY-1(02D) were conducted, including radiometric calibration, atmospheric correction, alignment, and cropping. Two sets of data, Liao Ning-01 (abbreviated as LN01) and Liao Ning-02 (abbreviated as LN02), were obtained for fusion experiments. We also acquired the corresponding classification ground truth (10 m) (available at https://drive.google.com/drive/folders/1JLCCB6ld5R49HDLN5SsMISx1d0fuqRjO).

LN01: As shown in Fig. 6, the area is located in the northwest of Jinpu New District, Dalian City, Liaoning Province, with coordinates of 121°50.49E and 39°21.98N. The LrHSI dimension of these data is (300, 300, 166) and the HrMSI dimension is (900, 900, 8). The landcover categories in this area include reservoir, seawater, sandy soil, broken bridge, barren grass, bare soil, highway, railway, mountain vegetation, and arable land.

LN02: As shown in Fig. 6, the area is located in the Liao River Estuary wetland, Panjin City, Liaoning Province, with coordinates of 121°46.41E and 40°56.17N. The LrHSI dimension of these data is (512, 512, 144) and the HrMSI dimension is (1536, 1536, 8). The landcover categories of this area include Liao River, phragmites australis, paddy fields, suaeda salsa, intertidal muds, reed pond, construction land, aquaculture ponds, and industrial land.

B. Comparison Method

In order to evaluate the effectiveness of the proposed methods, state-of-the-art methods are selected for comparison

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(a) (b) (c) (d) (e) (f)

Fig. 6. LN01 and LN02 datasets and the corresponding 10-m spatial resolution GT. (a) LN01 HrMSI, (b) LN01 LrHSI, (c) LN01 10 m GT, (d) LN02 HrMSI, (e) LN02 LrHSI, and (f) LN02 10-m GT.

using the same dataset in this article. The comparison methods include traditional methods such as GSA [26], FUSE [42], CNMF [20], and IR-TenSR [43]. There have also been unsupervised deep learning-based fusion methods developed in recent years, such as SURE [44], SURE-ERGAS [45], UDALN [38], UMC2FF [46], and M2U-Net [47]. Meanwhile, to demonstrate the role of the GCE module in the proposed fusion model, this article removes the GCE module from the model for experimental ablation comparison.

C. Quality Metrics for Simulated Data

In order to thoroughly compare the performance of different fusion methods, this article uses six metrics to evaluate the fusion results. These include the spectral angle mapper (SAM), the peak signal-to-noise ratio (PSNR), the relative dimensionless global error in synthesis (ERGAS), $Q2^n$, the root-mean-square error (RMSE), and the structural similarity index (SSIM).

With A and B representing the fusion result and the reference image, respectively, the SAM metric is usually used to measure the similarity between the fusion result's spectra and the reference image's spectra, which is defined as

$$\mathrm{SAM}(A,B)=\frac{1}{M}\sum_{j=1}^{M}\arccos\frac{A_j\cdot B_j}{\|A_j\|2\,\|B_j\|2} \tag{13}$$

where M represents the number of image elements of the spectrum and $\cdot$ denotes the inner product of two vectors. The smaller SAM indicates the better fusion effect.

The RMSE is a relatively common evaluation metric, which is defined as

$$\mathrm{RMSE}(A,B)=\sqrt{\frac{\|A-B\|_F^2}{N}} \tag{14}$$

where N represents the number of image elements.

The PSNR is used to evaluate the quality of the fused image and the extent to which the original information is maintained and is calculated as follows:

$$\mathrm{PSNR}(A,B)=20\log\frac{255}{\mathrm{RMSE}(A,B)}. \tag{15}$$

The ERGAS mainly considers the tradeoff between the spatial resolution and spectral fidelity of the fused image, which is calculated as follows:

$$\mathrm{ERGAS}(A,B)=\frac{100}{d}\sqrt{\frac{1}{S}\sum_{i=1}^{S}\left(\frac{\mathrm{RMSE}(A_i,B_i)}{\mu(B_i)}\right)^2} \tag{16}$$

where d is the spatial subsampling factor and μ denotes the image means.

The SSIM compares the similarity between features such as structure, brightness, and contrast of an image. It can effectively assess the degree of difference between the fusion image and the reference image. This is calculated using the following formula:

$$\mathrm{SSIM}(A,B)=\frac{1}{E}\sum_{i=1}^{E}\frac{\left(2\mu_A^i\mu_B^i\right)\left(2\sigma_{AB}^{ii}+C_2\right)}{\left(\mu^{2i}_A+\mu^{2i}_B+C_1\right)\left(\sigma^{2i}_A+\sigma^{2i}_B+C_2\right)} \tag{17}$$

where E denotes the number of image bands; μ_A^i and μ_B^i represent the mean values of A and B, respectively; σ^{2i}_A and σ^{2i}_B represent the variances of A and B, respectively; σ_{AB}^{ii} represents the covariance between A and B; and C_1 and C_2 are two constants.

$Q2^n$ is an image quality assessment index for LrHSI that utilizes hypercomplex numbers and is an extension of the $Q4$ index [48]. It is a generalization of the universal image quality index (UIQI) [49]. The UIQI calculates the similarity between a reference image band (x) and its corresponding target image band (y), which is calculated as follows:

$$Q(x,y)=\frac{4\sigma_{xy}\bar{x}\bar{y}}{(\sigma_x^2+\sigma_y^2)(\bar{x}^2+\bar{y}^2)'} \tag{18}$$

where $\bar{x}$ and $\bar{y}$ are the mean value of x and y, respectively; σ_x^2 and σ_y^2 are the variances of x and y, respectively; and σ_{xy} is the covariance between x and y.

The UIQI can be defined as

$$Q(x,y)=\frac{\sigma_{xy}}{\sigma_x\sigma_y}\frac{2\bar{x}\bar{y}}{\bar{x}^2+\bar{y}^2}\frac{2\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}. \tag{19}$$

The UIQI measures correlation, luminance distortion, and contrast distortion in monochromatic images, while the $Q4$ index models each pixel vector as a quaternion to overcome this limitation and account for spectral distortion in four-band images. $Q2^n$ extends the $Q4$ index by modeling each pixel spectrum as a hypercomplex number, specifically a 2n-ons. The calculation for a 2n-ons is as follows:

$$\mathbf{x}_j=x_{j,0}+x_{j,1}i_1+x_{j,2}i_2+\cdots+x_{j,2}^{n-1}i_2^{n-1}. \tag{20}$$

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TABLE I
QUANTITATIVE EVALUATION OF DIFFERENT METHODOLOGICAL FUSION APPROACHES AND ABLATION EXPERIMENT ON THE HT18 DATASET. THE BEST RESULTS ARE BOLDED

Method	SAM	PSNR	EGRAS	$Q2^n$	RMSE	SSIM
GSA	1.1106	49.4384	0.39323	0.99548	0.003153	0.99711
FUSE	1.6770	40.5931	0.91170	0.97590	0.007218	0.99221
CNMF	1.2217	43.6212	0.62514	0.98853	0.004852	0.99606
IR-TenSR	1.7786	44.0560	0.77606	0.98889	0.005682	0.99154
SURE	1.5700	43.8920	0.62733	0.99298	0.004597	0.99431
SURE-ERGAS	1.2725	46.5277	0.48189	0.99348	0.003681	0.99607
UDALN	1.0237	41.2212	0.76382	0.97002	0.006222	0.99682
UMC2FF	1.1892	49.6708	0.42799	0.99645	0.003455	0.99643
M2U-Net	1.0335	49.0129	0.40849	0.99547	0.003117	0.99689
UDS^2C^2\GCE	**0.8578**	42.9807	0.64001	0.98789	0.005117	0.99761
UDS^2C^2	0.8715	**52.2907**	**0.34183**	**0.99654**	**0.002597**	**0.99795**

D. Quality Metrics for Real Data

For real onboard data, this article uses a classification application-oriented evaluation. Specifically, the evaluation metrics for the classification results are overall accuracy (OA) and class-specific accuracy (CA); among them, OA is the accuracy of the classifier on all classes, which is calculated as the ratio of the number of correctly classified samples to the overall number of samples, and the specific formula is as follows:

$$\mathrm{OA} = \sum_{i=1}^{K} Y_i \Big/ \sum_{i=1}^{K} N_i \tag{21}$$

where Y_i denotes the number of categories i correctly categorized, N_i denotes the total number of samples in category i, and K is the total number of categories.

CA denotes the accuracy with which each category is correctly categorized and specifically can be expressed as

$$\mathrm{CA} = Y_i / N_i. \tag{22}$$

E. Results on the Simulated Data

Due to the existence of reference image on simulated data, this article first compares the fusion methods by calculating the metrics between the fusion results and the validation results on Ht18. The results of the evaluation metrics of the classical traditional methods and the state-of-the-art unsupervised deep learning fusion methods quantitatively on the Ht18 dataset are shown in Table I. The best result has been bolded to show that the model proposed in this article achieves excellent results in almost all metrics. Among them, the Ours\GCE is the fusion model proposed in this article by removing the GCE module as a way to demonstrate the role of the GCE module. As can be seen from the experimental performance, the model with the addition of the GCE module has a significant effect on the metric PSNR, and the performance on SAM is almost the same.

In addition, this article uses the same evaluation index to evaluate the enhanced spectral–spatial downsampled model. Since the input image of the simulated data is artificially degraded, the reference image of the LrMSI can be further artificially degraded in the same way. This reference image is then used as a validation to evaluate the performance of spectral–spatial downsampled models. The specific quantitative comparisons are shown in Table II. Our\GS is based on the removal of GCE and the replacement of SCFConv with normal static convolution. It shows an improvement of approximately 3% in PSNR by replacing the convolution kernel with SCFConv without modifying the model. In addition, the addition of the GCE module greatly improves the spectral–spatial downsampled model. This improvement is mainly attributed to the fact that the GCE module further eliminates radiometric difference between images to be fused and achieves better degradation effects.

To provide a visual and qualitative comparison, Fig. 7 displays the ground truth (GT) and true color images of fusion results obtained by different fusion methods on the Ht18 dataset. The first and third rows show the GT and true color images, respectively, while the second and fourth rows display the GT and residual plots for each result, respectively. It is obvious that the method proposed in this article closely matches the GT residual plots, demonstrating optimal recovery of spatial and spectral details and a good fusion performance.

F. Results on the Real Data

Figs. 8 and 9 show the results of the HrHSI obtained using the fusion proposed method for LN01 and LN02, respectively. The fusion results are basically the same color as the input data, indicating that no spectral distortion occurs in the results. The spatial texture features also maintain a good effect. In order to evaluate the model's performance more intuitively, we compared the spectra of typical features in the fusion results of LN01 and LN02 with their corresponding LrHSI. The line graph shows that the spectral curves of the features in the two images are largely similar, suggesting that the proposed method achieves a better spectral reconstruction effect.

In order to quantitatively evaluate the performance of different fusion methods on real data, this article applies the traditional method FUSE and the unsupervised methods

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TABLE II
QUANTITATIVE PERFORMANCE OF SCFCONV AND GCE MODULES ON THE ROLE OF SPECTRAL–SPATIAL DOWNSAMPLED MODEL. THE BEST RESULTS ARE BOLDED

Method	SAM	PSNR	EGRAS	$Q2^n$	RMSE	SSIM
UDS^2C^2\GS	0.0918	41.1129	0.85497	0.97432	0.0064831	0.99900
UDS^2C^2\GCE	**0.0052**	44.6445	0.56291	0.97526	0.0043406	0.99953
UDS^2C^2	0.0615	**72.3351**	**0.03326**	**0.99981**	**0.0002595**	**1**

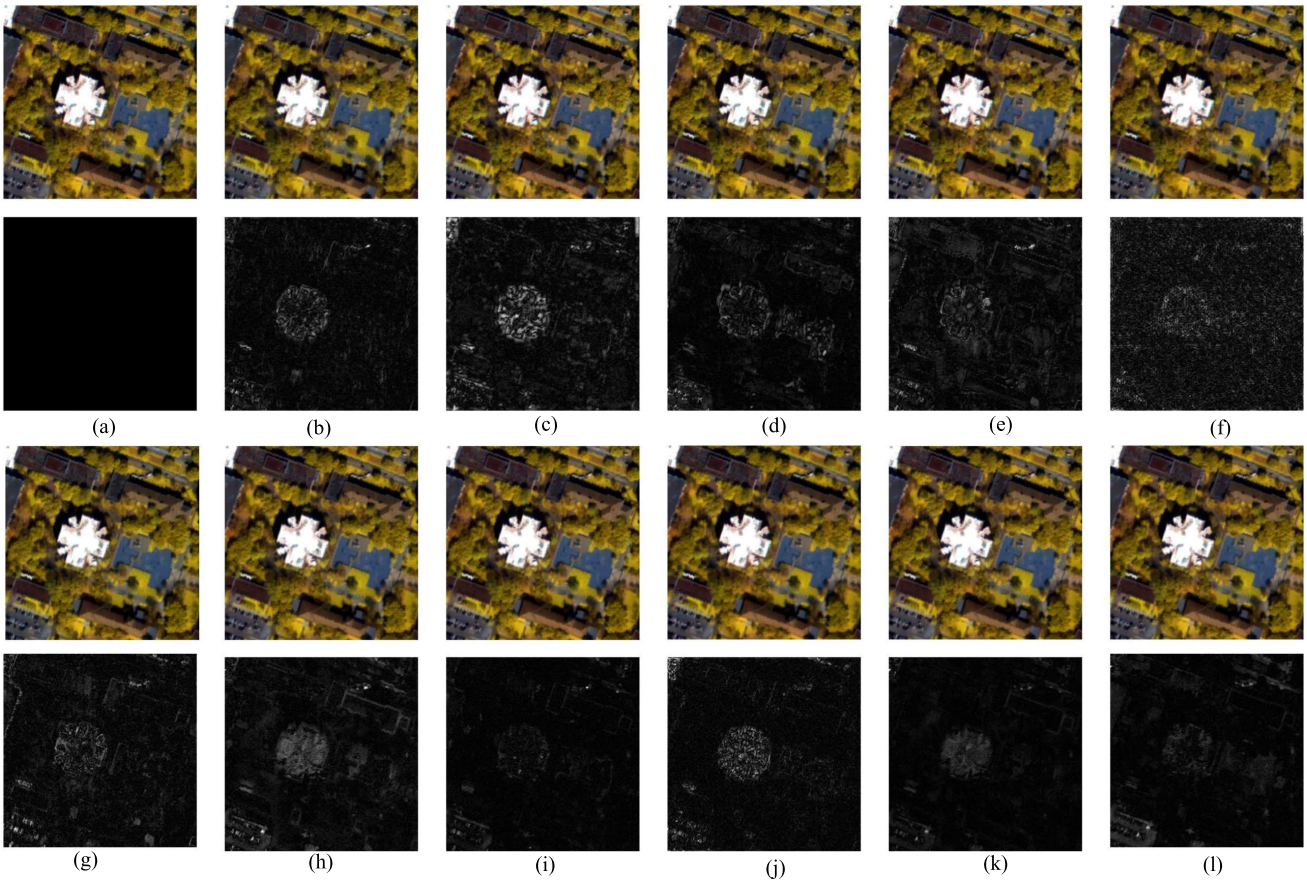


Fig. 7. First and third rows show the true color fusion results of the Ht18 dataset under different methods. The second and fourth rows show the residuals corresponding to each fusion result. (a) GT, (b) GSA, (c) FUSE, (d) CNMF, (e) SURE, (f) IR-TenSR, (g) SURE-ERGAS, (h) UDALN, (i) UMC2FF, (j) M2U-Net, (k) UDS^2-C^2\GCE, and (l) UDS^2C^2.

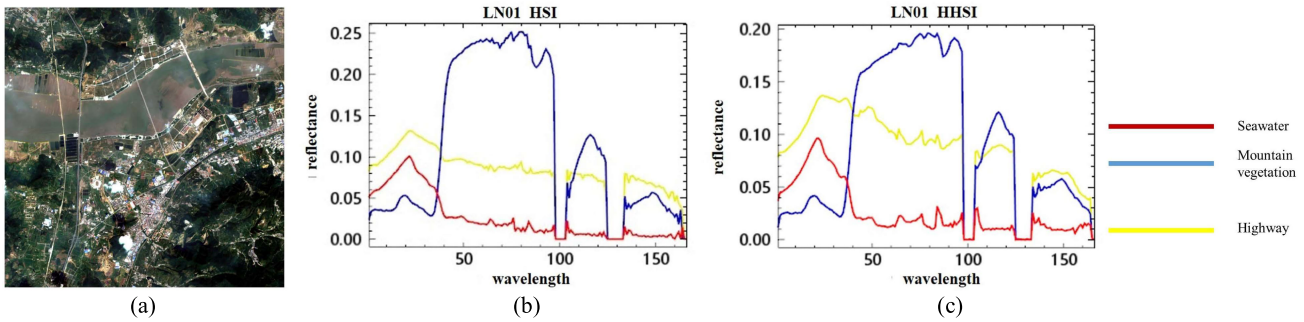


Fig. 8. LN01 fusion results and spectral comparison with LrHSI. (a) LN01 fusion result (HrHSI), (b) LN01 LrHSI typic feature spectra, and (c) LN01 HrHSI typic feature spectra.

UDALN and UDS^2C^2 to the real data LN01 and LN02 to obtain the corresponding fusion results. In this article, the two sets of real data and their fusion results are compared and analyzed using the deep learning-based classification method 3DCNN [50] and the traditional classification method random forest [51], respectively. In addition, the reference classification GT for LrHSI is obtained by downsampling the high spatial resolution GT.

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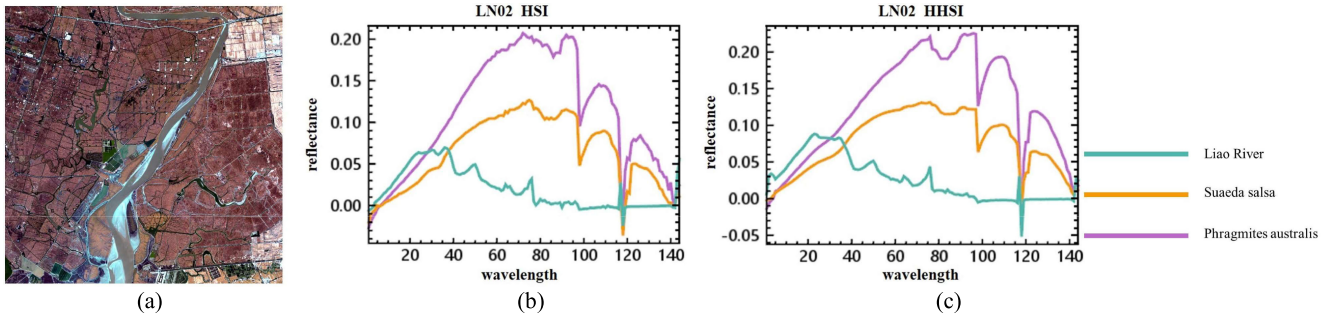


Fig. 9. LN02 fusion results and spectral comparison with LrHSI. (a) LN02 fusion result (HrHSI), (b) LN02 LrHSI typic feature spectra, and (c) LN02 HrHSI typic feature spectra.

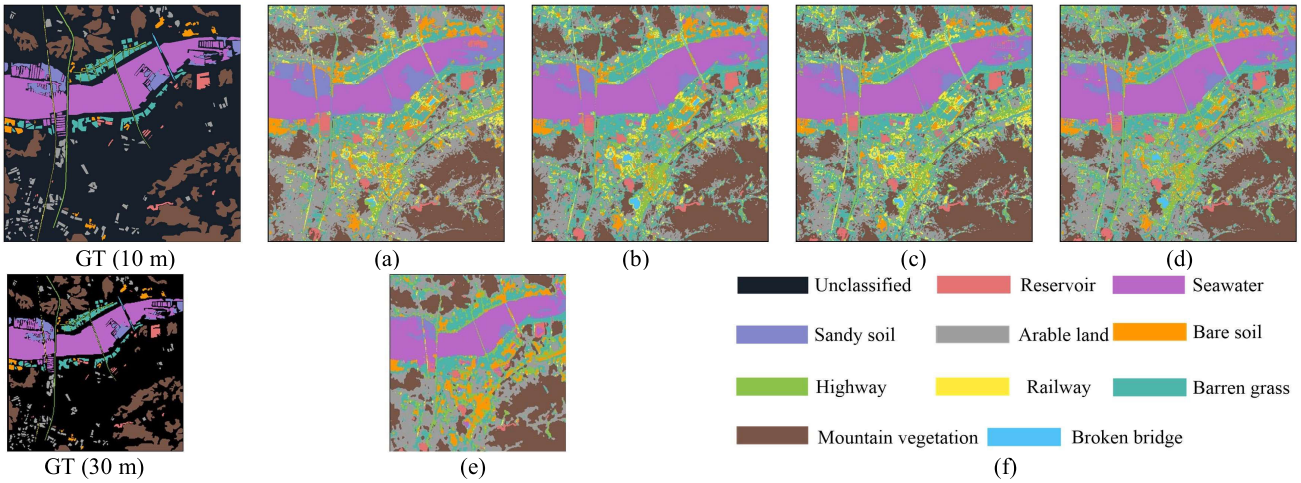


Fig. 10. 10- and 30-m spatial resolution LN01 GT, classification results of LN01 fusion HrHSI, LrHSI, and HrMSI under 3DCNN. (a) HrMSI (88.68%), (b) HrHSI(FUSE) (92.69%), (c) HrHSI(UDALN) (93.13%), (d) HrHSI(UDS^{2}C^2) (**93.72%**), (e) LrHSI (89.78%), and (f) category information.

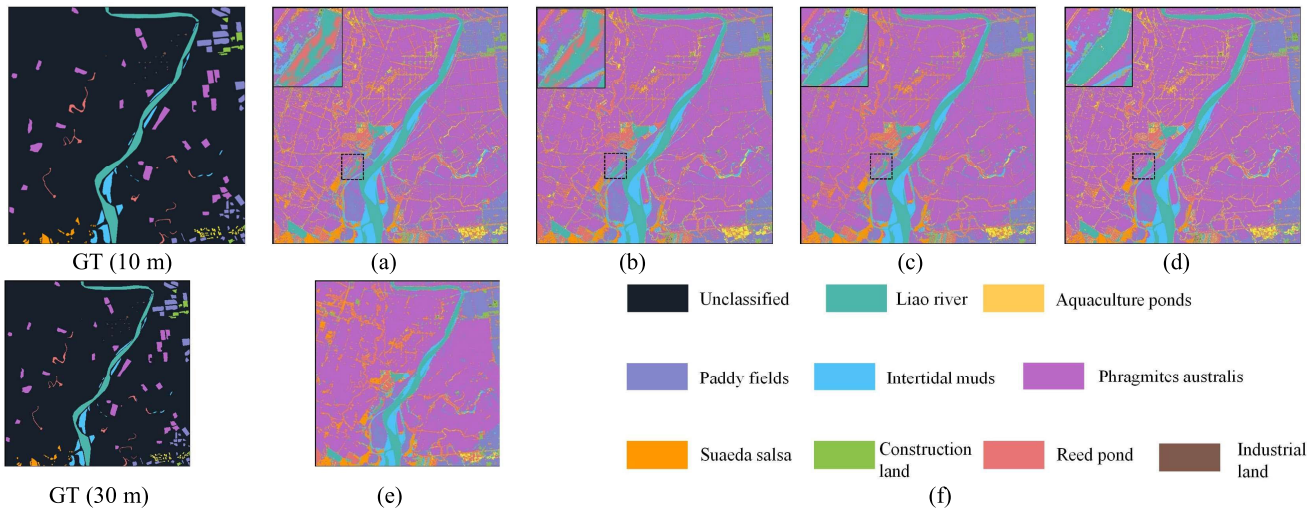


Fig. 11. 10- and 30-m spatial resolution LN02 GT, classification results of LN02 fusion HrHSI, LrHSI, and HrMSI under random forest. (a) HrMSI (96.27%), (b) HrHSI(FUSE) (96.71%), (c) HrHSI(UDALN) (96.91%), (d) HrHSI(UDS^{2}C^2) (**97.56%**), (e) LrHSI (96.37%), and (f) category information.

According to the characteristics of the sample size of the LN01 and LN02 datasets, 5% and 0.5% of the LN01-related data and LN02-related data were selected as training samples for the experiments, respectively. Finally, get the corresponding classification result map of the whole graph and the classification accuracy, as shown in Figs. 10 and 11, and Tables III and IV.

For the LN01 dataset, the following conditions hold.

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TABLE III
QUANTITATIVE CLASSIFICATION PERFORMANCE OF DIFFERENT COMPARISON DATA OF LN01 DATASET UNDER 3DCNN

Class	LN01_LrHSI	LN01_HrMSI	LN01_FUSE	LN01_UDALN	**LN01_UDS^{2}C^2**
Reservoir	72.32%	72.05%	79.18%	**79.77%**	76.56%
Seawater	96.06%	90.09%	96.68%	**96.76%**	96.63%
Sandy soil	81.11%	64.87%	83.15%	**85.33%**	85.03%
Broken bridge	47.50%	**94.43%**	91.15%	93.40%	93.59%
Barren grass	87.58%	86.00%	86.42%	86.35%	**88.00%**
Highway	76.39%	74.30%	79.07%	87.35%	**88.91%**
Railway	38.24%	62.63%	60.22%	69.12%	**70.24%**
Bare soil	80.00%	**85.24%**	82.51%	84.08%	85.04%
Mountain vegetation	93.96%	97.39%	96.97%	97.23%	**97.74%**
Arable land	**88.15%**	85.88%	79.74%	79.84%	85.81%
OA	89.78%	88.68%	92.69%	93.13%	**93.72%**

TABLE IV
QUANTITATIVE CLASSIFICATION PERFORMANCE OF DIFFERENT COMPARISON DATA OF LN02 DATASET UNDER RANDOM FOREST

Class	LN02_LrHSI	LN02_HrMSI	LN02_FUSE	LN02_UDALN	**LN02_UDS^{2}C^2**
Liao river	88.13%	95.43%	95.54%	93.92%	**95.60%**
Phragmites australis	98.55%	97.70%	97.72%	97.62%	**98.75%**
Paddy fields	97.11%	94.89%	95.77%	96.20%	**97.59%**
Intertidal muds	97.74%	97.10%	97.60%	**98.41%**	98.12%
Suaeda salsa	99.08%	**99.76%**	99.63%	99.73%	99.67%
Construction land	**86.70%**	73.86%	81.21%	84.28%	85.26%
Industrial land	57.10%	85.20%	83.54%	83.99%	**85.33%**
Aquaculture ponds	92.08%	94.18%	93.20%	**94.27%**	93.10%
Reed pond	21.27%	18.83%	26.12%	31.94%	**45.54%**
OA	96.37%	96.27%	96.71%	96.91%	**97.56%**

1) LN01 LrHSI and HrMSI achieved satisfactory classification performance, proving the viability of the dataset and GT produced by our team.
2) However, LrHSI's classification result map is blurred due to low spatial resolution, and HrMSI's classification accuracy is inferior to other fusion results due to insufficient spectral information.
3) By comparison of different fusion results, the proposed method achieved advantages in overall accuracy, especially on categories such as barren grass, highway, railway, and mountain vegetation, which proved the effectiveness of this method on real data.

For the LN02 dataset, similar to the classification results of LN01 dataset, the proposed method holds the advantage on the overall accuracy, especially on categories such as Liao River, phragmites australis, paddy fields, and reed pond. In addition, as shown in Fig. 11, local zooming of each classification map reveals that the UDS^{2}C^2 (HrHSI) classification results are more consistent with the real feature distribution.

VI. CONCLUSION

In this article, a UDS^{2}C^2 model is proposed for the hyperspectral image fusion task. The main contribution of this method is to design a 3-D dynamic convolution kernel by introducing an attention mechanism, and it can improve the overall feature extraction capability of the model. In addition, four modules were designed for the fusion task, and ablation experiments were designed to prove the role of the grouped convolution enhancement (GCE) module. The GCE module greatly improved the degradation performance by reducing the radiometric difference between images. Finally, due to the lack of evaluation methods for real data, this article designed an evaluation experiment of fusion methods for classification applications. This method has achieved excellent performance on both simulated data and real data, which proves the generalization of the proposed method.

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